

Weighted Filter Penalty Prediction for QoT Estimation

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Abstract: We experimentally show that noise distribution with regards to filters impacts the QoT by several dB. We thus propose a new QoT prediction method achieving up to 1.7dB margin reduction for links with high hop counts.

OCIS codes: 060.1155 All-optical networks; 230.7408 Wavelength filtering devices; 060.1660 Coherent communications

1. Introduction

The ever-increasing capacity needs of optical communication networks push metro/regional networks to evolve towards spectrally-efficient links, reasonably targeting 100G to 400G channel capacities with 37.5GHz spacing [1]. Yet, these capacity upgrades may be restrained by filter impairments, due to the typically high hop count of metro links [2], hence the importance of accurately predicting filter penalties to optimize metro networks.

Up to recently, empirical models were accurately estimating filter penalties up to 1dB, for instance predicting 0.03dB/filter for a typical 32.5GBd DP-QPSK with 50GHz spacing [3]. However, we showed in a recent paper [4] that such methods were unreliable for filter penalties beyond 1dB, when they can exceed 3dB for commercial 200G links with 37.5GHz spacing [5]. Furthermore, Bertran-Pardo *et al* [5] experimentally showed that the filter penalty of a commercial three-filter cascade roughly varies from 1.7dB to 3.2dB when noise loading is moved from before to after the filter cascade. This clearly shows it is necessary to account for lightpath topology, more specifically for noise distribution across the filter cascade, to accurately account for filter penalties, and thus to reduce margins.

In this paper, we first study the maximal range of values the filter penalty can take depending on filter bandwidth and OSNR level on a custom experimental setup. We then describe a new QoT prediction method that weights best-case and worst-case filter penalties according to lightpath (LP) topology. Finally, we numerically assess the accuracy of this new method targeting spectrally-efficient, high hop count links exhibiting large filter penalties.

2. Experimental investigation of the variation range of filter penalties

As shown in [5], lowest filter penalties are observed when all noises are loaded before a filter cascade. Conversely, highest penalties occur when all noises are loaded before a filter cascade. Indeed, the former, “best-case” (BC) configuration filters signal and noise equally when the latter, “worst-case” (WC) configuration only filters the signal. The two configurations thus provide the range of values the filter penalty can take for a given filter cascade, whatever the LP topology. The experimental testbed reproducing these two configurations is depicted in Fig.1(a).

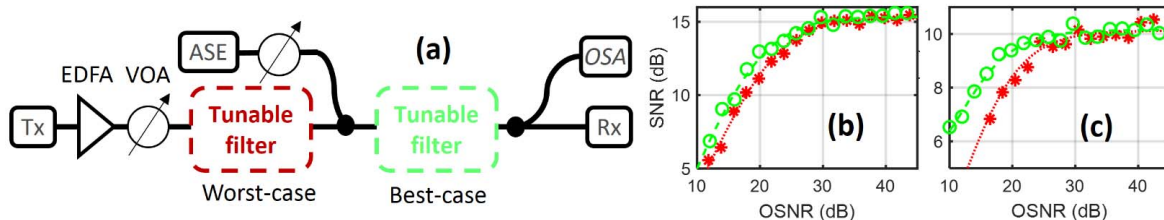


Fig. 1. (a) Experimental setups in “worst-case” and “best-case” configurations. Measured SNR vs OSNR curves for both configurations, for a DP-QPSK at 32.5GBd with RRC0.1 pulse shape with an optical filter with 38GHz (b) and 22GHz (c) bandwidths.

Both transmitter (Tx) and receiver (Rx), including offline digital signal processing (DSP), are the same as described in [4]. For noise loading, we use a commercial ASE noise source coupled with a variable optical attenuator (VOA) to control the amount of loaded noise. On Rx side, we shunt 1% of the signal into an optical spectrum analyzer (OSA) to quantify signal and noise levels, from which we derive OSNR values. In Fig.1(b-c), we plot SNR-OSNR curves [4] obtained for a RRC pulse-shaped 32.5GBd DP-QPSK with 0.1 roll-off crossing a tunable filter with bandwidths of 38GHz and 22GHz, respectively. In qualitative agreement with [5], we experimentally show the filter penalty is roughly within a 2dB range at critical forward error correction (FEC) threshold for the tighter filter. For the larger filter, the difference between BC and WC curves is lower but still significant, up to 1dB.

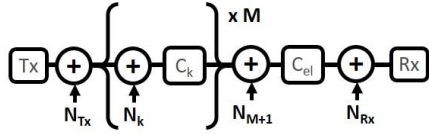
3. Weighted filter penalty prediction method

We model a typical lightpath as depicted in Fig.2. We account for transmitter (Tx) and receiver (Rx) noises as additive white gaussian noise (AWGN) sources N_{Tx} and N_{Rx} . The total number of crossed optical filters, written C_k ,

is M . We introduce $(M+1)$ AWGN noise sources N_k to account for all propagation noises occurring before (N_1), after (N_{M+1}) and inside ($N_k, k=2$ to M) the filter cascade. We emulate the low-pass filtering behavior from Rx optoelectronics through an additional filter function C_{el} . We propose to predict the filter penalty based on the assumption that, for identical OSNR levels, the SNR of the lightpath (LP) depicted in Fig.2 can be written as:

$$SNR = w \cdot SNR_{BC} + (1-w) \cdot SNR_{WC} \quad (1)$$

where w is a LP-dependent weight and $SNR_{BC/WC}$, are the SNRs obtained in BC and WC configurations, respectively. The weight w is deduced from Eq.1-2 where Σ appears as a sum of all AWGNs weighted by their



$$w = \frac{\Sigma - \Sigma_{BC}}{\Sigma_{WC} - \Sigma_{BC}} \quad (2) \quad \Sigma = N_{Tx} + \sum_{p=1}^{M+1} p \cdot N_p + (M+2) \cdot N_{Rx} \quad (3)$$

$$\Sigma_{WC} = N_{Tx} + (M+1) \sum_{p=1}^{M+1} N_p + (M+2) N_{Rx} \quad (4) \quad \Sigma_{BC} = N_{Tx} + \sum_{p=1}^{M+1} N_p + N_{Rx} \quad (5)$$

Fig. 2. Model for a typical lightpath | Equations for weight calculations

positions along the filter cascade, C_{el} included. The sum Σ thus characterizes how AWGNs are distributed along the lightpath, and Σ_{WC} (Eq.4) and Σ_{BC} (Eq.5) correspond to “worst-case” and “best-case” configurations, respectively. Whereas Σ_{WC} is a direct application of Eq.1, Σ_{BC} is deliberately underestimated by assuming that Rx noise is filtered by the full cascade. This achieves a slight and systematic overestimation of the filter penalty, making the prediction compatible with a safe definition of margins: with $0 < w \leq 1$, we have $SNR_{WC} \leq SNR < SNR_{BC}$. We use the following expression to characterize the SNR-OSNR relation [4]:

$$SNR^{-1} = \sum_{n=0}^{n=2} a_{n,BC/WC} (R/B)^n OSNR^{-n} \quad (6)$$

where R is the baudrate, B is the reference band for OSNR measurement and $a_{n,BC/WC}$ are calibrated parameters. The weighted prediction method works as following. For calibration, we perform back-to-back experiments as in section 2 to obtain SNR-OSNR curves for various amounts M of cascaded filters, for both WC and BC configurations. We then fit experimental SNR-OSNR curves using Eq.6 to obtain $(a_{n,BC/WC} \text{ vs } M)$ calibration curves. Then, for any LP:

- 1) Using the GN model [6], we calculate the generalized OSNR value as well as AWGN powers $N_{Tx/Rx}$ and N_k .
- 2) We interpolate $a_{n,BC/WC}$ values according to the amount M_{LP} of filters crossed.
- 3) Inputting $a_{n,BC/WC}$ and OSNR in Eq.6, we deduce SNR_{BC} and SNR_{WC} values.
- 4) Inputting AWGN powers in Eqs.3-5, we calculate w with Eq.2, then SNR with Eq.1.

4. Numerical validation of the filter penalty prediction method

We numerically simulate optical DP-QPSK, DP-16QAM and DP-32QAM transmissions, including BC and WC calibrations, using the model depicted in Fig.2. We set N_{Tx} and N_{Rx} equal, and so that back-to-back performance is similar to experimental results obtained in section 2. We generate random sequences of 2^{16} symbols with 24 samples per symbol (SPS). At Tx output, signals are RRC-pulse shaped with 0.1 roll-off. Optical filters C_k traversed by the signal, based on the 37.5GHz grid, are defined using a realistic WSS model [7] with state-of-the-art sharpness, and are aligned with Tx and Rx lasers. At Rx input, the signal crosses an electrical filter C_{el} calibrated from the experimental setup (Fig.1). After sampling at 2 SPS, the signal is equalized using a typical CMA with 25 taps.

We first assess the validity of the prediction method when the noise is evenly distributed in between optical filters. In Fig.3(a), we plot the SNR penalty ΔSNR , i.e. the filter-induced SNR degradation at constant OSNR, for a DP-QPSK with $M=10$. Similarly to our experimental study (cf. Fig.1), there is a significant difference between WC and BC configurations, of about 2dB for OSNRs between 15 and 20dB. We further verify the curve obtained when

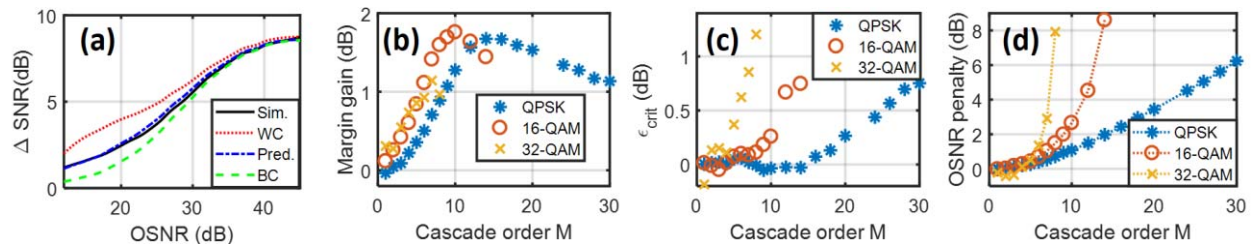


Fig. 3. For evenly distributed noise in between optical filters, (a) ΔSNR for a DP-QPSK with $M=10$. For various modulation formats and as function of M , (b) the OSNR penalty (c) the critical error ϵ_{crit} and (d) the extra OSNR margin the method is achieving.

noise is evenly distributed (Sim.) is bounded by BC and WC curves, and closely matches the prediction (Pred.), with a maximum error of 0.1dB over the whole OSNR range. Fig.3(a) also illustrates the strong variation of Δ SNR depending on the OSNR value. For fair comparison between modulation formats, we thus focus on the OSNR penalties Δ OSNR (cf. Fig.3b) i.e. the extra OSNR requirement due to the optical filter cascade to achieve a critical SNR value corresponding to a pre-FEC BER threshold of 1.7×10^{-2} : 6.5dB for DP-QPSK, 13dB for DP-16QAM and 15.9dB for DP-32QAM. We then assess the method via the critical error ϵ_{crit} defined as the difference between predicted and simulated OSNR penalties. From Fig.3(c), the proposed method achieves critical errors under 0.1dB up to 15 cascaded filters for DP-QPSK. The error regularly increases for larger cascades, but does not exceed 0.8dB for a 30-filter cascade presenting an OSNR penalty over 6dB. For DP-16QAM, the critical errors remain below 0.8dB for cascades up to 14 filters (8.2dB OSNR penalty) and for 32-QAM, critical errors remain below 1.2 dB for cascades up to 8 filters (8dB OSNR penalty). For OSNR penalties below 4dB, the critical errors remain below 0.5dB whatever the modulation format. More importantly, all prediction errors above 0.1dB are overestimation of the penalty. This property, due to the biased estimation of w through Σ_{BC} (Eq.2-4), has a cost in terms of absolute prediction accuracy. However, it defines the predicted penalty as an upper bound of the actual penalty, less pessimistic than the “worst-case” scenario where all noise is loaded after the filter cascade. We illustrate this in Fig.3(d) by plotting the margin gain, i.e. the reduction of the required OSNR when our prediction method is used, compared to when the “worst-case” penalty is assumed. For low hop counts, the gain of our prediction method is limited. However, results show that for lightpaths with high hop counts, the gain can reach 1.7dB for DP-QPSK and DP-16QAM, and 1.2dB for DP-32QAM.

Next, we assess the robustness of the prediction method when the noise is not evenly distributed between the filters. To do so, we proceed to Monte-Carlo simulations where, at each repetition, the noise is randomly distributed in between a cascade of six 37.5GHz filters for a DP-16QAM transmission. Results are gathered in Fig.4.

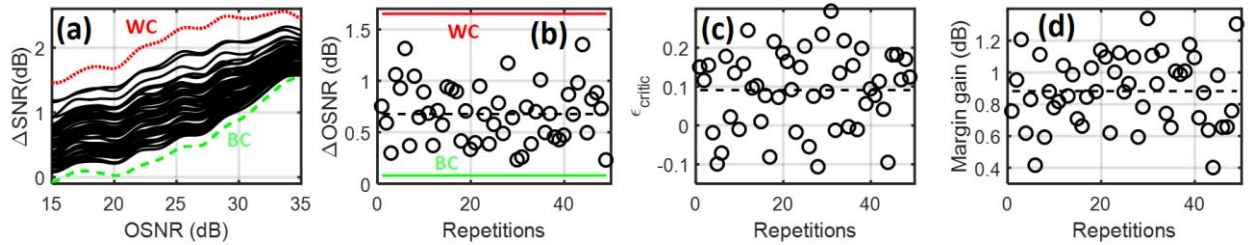


Fig. 4. Simulated (a) Δ SNR as the OSNR-dependent SNR penalty, (b) Δ OSNR as the OSNR penalty at 13dB of SNR, (c) OSNR penalty error ϵ_{crit} and (d) extra OSNR margin for all 50 repetitions with random distribution of the noise in between six 37.5GHz-filters for a DP-16QAM.

In Fig.4(a), we plot the various simulated Δ SNR curves for the 50 Monte-Carlo repetitions, each associated to a signal distribution of noise in-between the optical filters. We observe that all simulated values are well distributed within the upper and lower bounds defined by BC and WC configurations, respectively. In Fig.4(b), we plot Δ OSNR obtained for all repetitions. We show here that for the same filter cascade, the OSNR penalty varies from 0.2 to 1.4dB depending on how the noise is distributed along the lightpath, with a mean value around 0.7dB. In Fig.4(c), the critical prediction error ϵ_{crit} fluctuates between -0.1 and 0.3dB with a mean value around 0.1dB. We count approximately 20% of negative values correspond to penalty underestimations. However, those remain small compared to the OSNR penalty, and, as shown in Fig.3(c), do not occur for larger filter penalties. Finally, we also confirm that for unevenly distributed noise, there is still a significant gain in using our prediction method, of 0.9dB in average when compared to worst-case dimensioning.

5. Conclusion

We experimentally demonstrated the significant impact of the relative placement of noise sources and optical filters when it comes to the filter penalty due to a WSS cascade. Based on this evidence, we proposed a semi-analytical method that slightly overestimate the filter penalty in most cases. We showed through numerical simulations that the method can achieve OSNR penalty predictions accurate within 0.1dB for filter penalties up to 3dB, and that gains of up to 1.7dB can be achieved for both DP-QPSK and DP-16QAM compared to worst-case dimensioning.

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6. References

- [1] A. Morea et al, JOCN 7(2), 293-300 (2015)
- [2] J. Pan et al, OFC proceedings, M2E.5 (2017)
- [3] J. Pesic et al, OFC proceedings, M3K.2 (2016)
- [4] C. Delezoide et al, ICTON proceedings, WE.B1.1 (2017)
- [5] O. Bertran-Pardo et al, OFC proceedings, M3A.3 (2015)
- [6] A. Carena et al, JLT 30(10), 1524-39 (2012)
- [7] C. Pulikkaseril et al, OE 19(9), 8458-70 (2011)